ML Report:
VoiceBridge

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1 Introduction

The VoiceBridge project addresses the digital accessibility gap faced by individuals with speech impairments resulting from neurological conditions like Parkinson's disease, stroke, or cerebral palsy. While modern Automatic Speech Recognition (ASR) models have advanced significantly, flagship architectures such as Whisper and wav2vec often exhibit drastically reduced performance when processing atypical or dysarthric speech. This creates significant barriers to independence, preventing users from performing everyday digital tasks like messaging or browsing the internet. The goal of this initiative is to bridge this gap by fine-tuning a machine learning model to reliably interpret impaired speech and map it to actionable digital commands.

The system's objective is to capture impaired audio through a standard microphone and process it via a custom-trained machine learning model. The resulting text output is not merely for transcription but is designed to be mapped into actionable commands. By transforming these inputs into device-level controls—such as sending messages or operating applications—VoiceBridge aims to restore autonomy to users in everyday environments, including homes and clinics. This objective heightens the need for an accurate model to provide transcripts, as commands are often only a couple words in length. Due to the lack of context available to the model when processing short phrases, high recognition accuracy and low word error rate (WER) are vital for the systems relying on these model's performance.

2 Related Work

Here, talk about the related work you encountered for your approach. Cite at least 5 references. Refer to item 2. No one has done exactly your task? Write about the most similar thing you can find. This should be around 0.25-0.5 pages.

3 Dataset

Training and evaluation data are drawn from the Speech Accessibility Project (SAP), a large-scale corpus of atypical speech collected to support ASR research for individuals with dysarthria and related conditions. SAP is well-suited to this work because it captures the acoustic characteristics, such as breathiness, irregular timing, and strained phonation, which cause performance degradation in general-purpose speech recognition systems.

A subset of 400 labeled utterances was used for fine-tuning, with an 80/20 train/validation split (320 and 80 samples respectively). A fully disjoint held-out set was reserved for evaluation; each utterance is paired with a ground-truth transcript and structured ratings covering the speaker's diagnosed condition and symptom severity across multiple dimensions. Training was conducted on SHARCNET H100 GPUs. As shown in Figure **??**, the subset is dominated by Parkinson's Disease (59.4%) and ALS (26.1%), with Cerebral Palsy (10.1%), Down Syndrome (2.6%), and Ataxic Dysarthria (1.8%) providing coverage of acoustically distinct profiles.

4 Features

The input representation follows the standard Whisper preprocessing pipeline: all audio is resampled to 16 kHz and converted to log-Mel spectrograms via `WhisperProcessor`. Domain adaptation is achieved through Low-Rank Adaptation (LoRA), which augments the frozen pretrained model with trainable low-rank matrices injected into the attention layers. Rather than modifying the base weights, LoRA allows the model to acquire task-specific adjustments within a compact parameter space, learning the acoustic characteristics of dysarthric and command-style speech. Adapters are applied to the query and value projection matri-

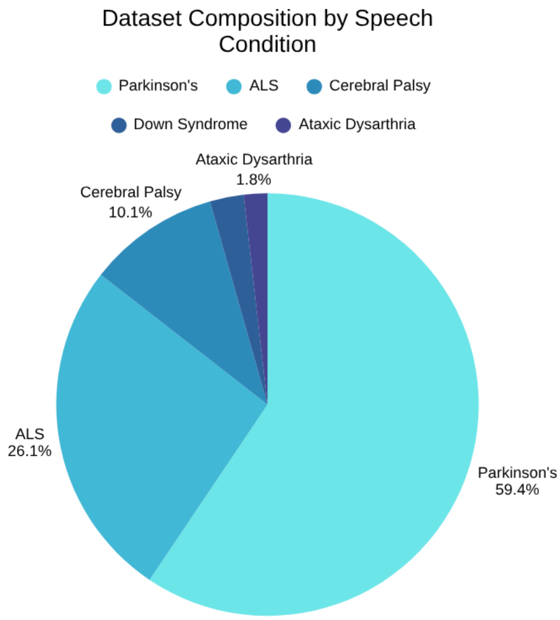

Figure 1: Condition distribution of the SAP training subset.

ces (q_proj, v_proj); training configuration and hyperparameters are detailed in Section ??.

5 Implementation

Describe your model and implementation here. Refer to item 4. This may take around a page.

6 Results and Evaluation

Transcription quality is measured using Word Error Rate (WER), defined as

$$\mathrm{WER} = \frac{S + D + I}{N},$$

where S, D, and I denote substitutions, deletions, and insertions, and N is the reference word count. For short commands, WER is a particularly consequential metric: a single error on a one- or two-word utterance constitutes a complete transcription failure and prevents command execution. All predictions and references are lowercased prior to scoring.

Baseline vs. fine-tuned. An unmodified `openai/whisper-small` model serves as the baseline. Both models are evaluated on the same 52-sample held-out set under identical decoding conditions. The baseline achieves an average WER of 0.813; the LoRA-adapted model reduces this to 0.558, an absolute reduction of 0.255 corresponding to approximately a 31% relative improvement across the test conditions.

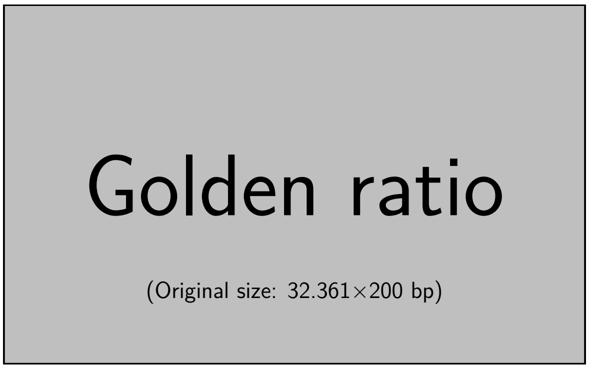

Figure 2: A figure with a caption that runs for more than one line. Example image is usually available through the `mwe` package without even mentioning it in the preamble.

Per-condition analysis. Performance improvements are distributed across diagnostic groups but vary in magnitude. For Parkinson's Disease and ALS, fine-tuning substantially increases the proportion of near-zero WER samples and reduces high-error outliers. For Cerebral Palsy, WER decreases from approximately 0.75 to 0.45; for Ataxic Dysarthria, from approximately 0.65 to 0.35. Symptom-level analysis confirms that error distributions shift toward lower values across severity levels, indicating more consistent performance rather than selective gains on lower-severity speakers.

7 Feedback and Plans

Write about your plans for the remainder of the project. This should include a discussion of the feedback you received from your TA, and how you plan to improve your approach. Reflect on your implementation and areas for improvement. Refer to item 6. This may be around 0.5 pages.

8 Template Notes

You can remove this section or comment it out, as it only contains instructions for how to use this template. You may use subsections in your document as you find appropriate.

8.1 Tables and figures

8.2 Citations

Many websites where you can find academic papers also allow you to export a bib file for citation or bib formatted entry. Copy this into the `custom.bib` and you will be able to cite the paper in the LaTeX. You can remove the example entries.

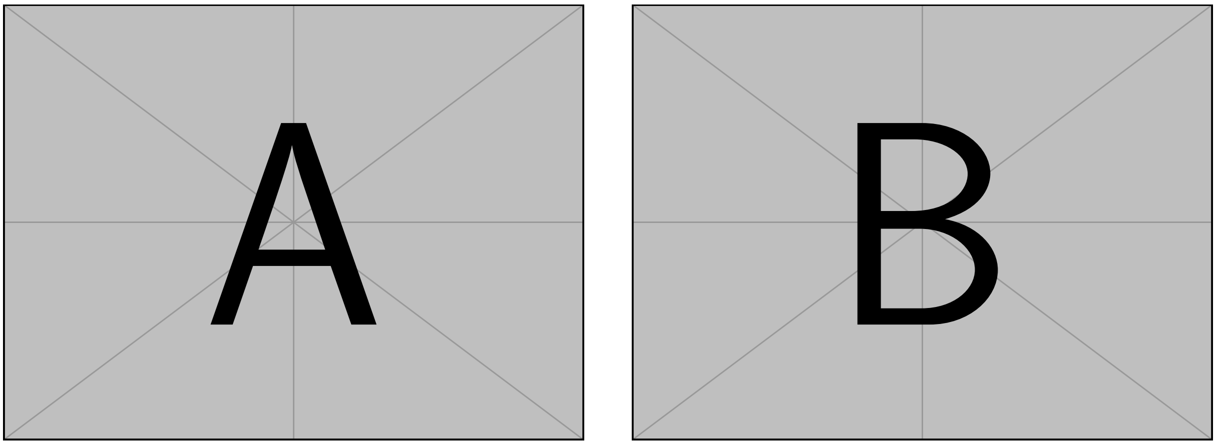

Figure 3: A minimal working example to demonstrate how to place two images side-by-side.

Output	natbib command	ACL only command
(**?**)	`\citep`	
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Table 1: Citation commands supported by the style file.

8.3 Equations

An example equation is shown below:

$$A = \pi r^2 \tag{1}$$

Labels for equation numbers, sections, subsections, figures and tables are all defined with the `\label{label}` command and cross references to them are made with the `\ref{label}` command. This an example cross-reference to Equation **??**. You can also write equations inline, like this: $A = \pi r^2$.

Team Contributions

Write in this section a few sentences describing the contributions of each team member. What did each member work on? Refer to item 7.